

Tide retrospective:

full-line J-function asymptotic for the quartic potential

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Abstract

A single-session Tide-loop excursion in the `laplace` seabed, formalising Step 2 of the grammar paper’s precursor sequence: the full-line J-function asymptotic for the single-component blow-up $K(w) = w^4/24$. Two theorems committed: a centred DCT theorem $t^{1/4} \int e^{-tw^4/24} (\pi(w) - \pi(0)) dw \rightarrow 0$ (Candidate X from the deliberation), and the user-facing form $t^{1/4} J(t) \rightarrow \pi(0) \cdot \frac{1}{2} \cdot 24^{1/4} \cdot \Gamma(1/4)$ as a 2-line corollary using `quartic_partition`. 280 lines of Lean over the existing seabed, zero `sorry`, single-session formalisation. The deliberation reshaped the target from the direct form (Candidate A) to the centred form (X) — less algebra in the statement, the user-facing form falls out — which Claude initially preferred but GPT correctly pushed harder for. The K1 worktree discipline (landed earlier today) gave clean filesystem isolation; no collisions with the four other concurrent Tide branches.

1 Setting

The grammar paper precursor sequence is a multi-tide programme aimed at building enough infrastructure in the `laplace` seabed to formalise the resolution-of-singularities argument that drives the grammar paper’s main theorem. The sequence’s published steps so far:

- *Step 1* (Tide 7): bounded-prior partition function for the quartic — $Z_a(t) := \int_{[-a,a]} e^{-tw^4/24} dw$, with positivity, tail bound, and the splitting infrastructure that subsequent precursor steps reuse.
- *Step 2-prime* (Tide 11+): bounded-prior with continuous test function. Branch-side closure has primitives plus a Lipschitz unnormalised estimate; the main-branch merge has only the primitives.

This Tide is the next link in that chain: drop the bounded-prior restriction and prove the *full-line* J-function asymptotic. Concretely, for the single-monomial potential $K(w) = w^4/24$ and a continuous bounded prior $\pi : \mathbb{R} \rightarrow \mathbb{R}$, prove

$$t^{1/4} \cdot J(t) := t^{1/4} \int_{\mathbb{R}} e^{-tw^4/24} \pi(w) dw \longrightarrow \pi(0) \cdot \frac{1}{2} \cdot 24^{1/4} \cdot \Gamma(1/4) \quad \text{as } t \rightarrow \infty.$$

The Step-2 deliberation¹ drafted three candidates spanning scope (k=2 only / generic k / bounded-support variant) and converged on a fourth, GPT-proposed reshape: the *centred* form $t^{1/4} \int e^{-tw^4/24} (\pi(w) - \pi(0)) dw \rightarrow 0$, with the user-facing direct form derived as a 2-line corollary using `quartic_partition` (Tide 7’s closed form for the no-prior partition).

¹See `projects/primer/tide-log/2026-05-07-tide-grammar-j-function.md` in the SRI repo for the full deliberation transcript and Candidates v1.

2 The thing formalised

For $K(w) = w^4/24$ and $\pi : \mathbb{R} \rightarrow \mathbb{R}$ continuous with $|\pi(w)| \leq M$ for some M ,

$$\lim_{t \rightarrow \infty} t^{1/4} \int_{\mathbb{R}} e^{-t w^4/24} (\pi(w) - \pi(0)) dw = 0,$$

and consequently

$$\lim_{t \rightarrow \infty} t^{1/4} \int_{\mathbb{R}} e^{-t w^4/24} \pi(w) dw = \pi(0) \cdot \frac{1}{2} \cdot 24^{1/4} \cdot \Gamma(1/4).$$

The Lean signatures, in namespace `Laplace.OneD`:

```
theorem quartic_jfunction_centered_tendsto_zero
  {prior : ℝ → ℝ} (hprior_cont : Continuous prior)
  {M : ℝ} (hprior_bd : x, |prior x| ≤ M) :
  Tendsto (fun t : ℝ =>
    t ^ ((1:ℝ)/4) *
    ∫ w : ℝ, Real.exp (-(t * w^4 / 24)) * (prior w - prior 0))
  atTop (ℕ 0)

theorem quartic_jfunction_asymptotic
  {prior : ℝ → ℝ} (hprior_cont : Continuous prior)
  {M : ℝ} (hprior_bd : x, |prior x| ≤ M) :
  Tendsto (fun t : ℝ =>
    t ^ ((1:ℝ)/4) *
    ∫ w : ℝ, Real.exp (-(t * w^4 / 24)) * prior w)
  atTop (ℕ (prior 0 * (1/2) * (24:ℝ) ^ ((1:ℝ)/4) *
    Real.Gamma (1/4)))
```

located at `Laplace/OneD/JFunctionQuartic.lean`.

The hypothesis ‘Continuous prior’ is slightly stronger than the deliberation’s ‘AEMeasurable + ContinuousAt 0 + bounded’. The Lean saving (50 lines of `AEMeasurable.comp_quasiMeasurePreserving` plumbing) was worth it for a Tide step; the weakening is recorded as a follow-up. All paper applications use continuous priors anyway.

3 Proof strategy

The proof is substitute-then-DCT. With $a := t^{1/4}$ (so $a^4 = t$), the change of variables $u = a \cdot w$ converts

$$t^{1/4} \int_{\mathbb{R}} e^{-t w^4/24} (\pi(w) - \pi(0)) dw$$

into the t -independent-integrand form

$$\int_{\mathbb{R}} e^{-u^4/24} (\pi(u/a) - \pi(0)) du.$$

The change of variables is implemented by Mathlib’s `MeasureTheory.Measure.integral_comp_mul_right`, which says $\int g(w \cdot a) dw = |a^{-1}| \cdot \int g(u) du$. Setting $g(u) := e^{-u^4/24} (\pi(u/a) - \pi(0))$, we have

$g(w \cdot a) = e^{-tw^4/24} (\pi(w) - \pi(0))$ (using $a^4 = t$ and $(w \cdot a)/a = w$), so the original integral equals $|a^{-1}| \cdot \int g(u) du = (1/a) \int g(u) du$. Multiplying by $a = t^{1/4}$ gives the substituted form.

The DCT step uses `MeasureTheory.tendsto_integral_filter_of_dominated_convergence` with envelope $2M e^{-u^4/24}$ (integrable by Tide 7’s `quartic_integrable_pow0`) and pointwise limit 0 (from $u/a \rightarrow 0$ as $a \rightarrow \infty$ plus continuity of π at the origin, then multiplication by the bounded $e^{-u^4/24}$).

The user-facing corollary follows by splitting $\pi(w) = (\pi(w) - \pi(0)) + \pi(0)$ inside the integral. The centred piece is the headline; the constant piece is $\pi(0) \cdot t^{1/4} \cdot \int e^{-tw^4/24} dw$, evaluated by Tide 7’s `quartic_partition` closed form. Multiplying out $t^{1/4} \cdot (24/t)^{1/4} = 24^{1/4}$ yields the constant $\pi(0) \cdot \frac{1}{2} \cdot 24^{1/4} \cdot \Gamma(1/4)$.

The proof body is structured as: a private substitution lemma `jfunction_centered_subst` (55 lines, the most arithmetically dense step—handles the `Real.rpow` Jacobian bookkeeping) and a small dominator- integrability helper, followed by the headline theorem (50 lines, DCT plus the substitution rewrite) and the corollary (70 lines, integral split + closed-form substitution).

4 Roadblocks and resolutions

‘ π ’ clashes with Lean 4.29’s namespace. The first skeleton bound the prior as $\{\pi : \mathbb{R} \rightarrow \mathbb{R}\}$. Lean rejected: unexpected token ‘ π ’; expected ‘ $_$ ’ or identifier. The Greek letter ‘ π ’ is somehow treated as an operator or reserved-ish in this Lean version. Rename to `prior`; the rebuild went through. Worth landing in `laplace/CLAUDE.md` as a v4.29 quirk.

`Real.tendsto_rpow_atTop` is at root namespace. First attempt at the pointwise limit step used `Real.tendsto_rpow_atTop`. Build error: Unknown constant ‘`Real.tendsto_rpow_atTop`’. The lemma lives at the root namespace as `tendsto_rpow_atTop` (in `Mathlib.Analysis.SpecialFunctions.Pow.Asymptotics`). This is the same gotcha pattern as `MeasureTheory.Measure.integral_comp_mul_right`—namespace inconsistency between morally-related lemmas. Worth a one-liner in `CLAUDE.md`.

‘`.symm`’ on “ doesn’t dot-resolve to ‘`Filter.EventuallyEq.symm`’. The combine-via-`Tendsto.congr` step originally wrote `Tendsto.congr h_decomp.symm h_sum_C`. Lean reported Invalid field notation: ... ‘`Filter.Eventually`’ is not a type that has a ‘`.symm`’ projection. The fix is `h_decomp.mono fun _ h => h.symm` (use `Filter.Eventually.mono` to symmetrise the inner equality pointwise) or explicit `Filter.EventuallyEq.symm h_decomp`. Subtle because in prose t in `atTop`, $f \ t = g \ t$ is $f = [\text{atTop}] \ g$ —but the `def` unfolding doesn’t propagate to `.symm` dot resolution. Worth a `CLAUDE.md` note.

Hypothesis weakening: `AEMeasurable` $\rightarrow$ `Continuous`. The deliberation chose `AEMeasurable + ContinuousAt 0 + global bound` as the mathematically-minimal hypothesis; the formalisation strengthened to `Continuous`. Reason: the `AEMeasurable` case needs `AEMeasurable.comp_quasiMeasurePreserving` for the composition $\pi \circ (\cdot/a)$, plus `AEStronglyMeasurable` follow-through, plus a similar argument inside the DCT bound step—collectively 50 lines of `MeasureTheory` plumbing. The strengthened hypothesis captures all paper applications and saves a Tide-substantial chunk of proof. The weakening is recorded as a follow-up tide; nothing in the existing proof obstructs the substitution.

No proof-side roadblocks beyond the above. The substitution lemma typechecked first try once the $a^4 = \text{tidentitywassetupcorrectly}(\text{thetrick} : \text{converttoReal.rpowviaReal.rpow_natCastthenuseReal.rpow_mul}). \text{TheDCTandcorollarywentthroughwithmod}$ — — nothrashing.

Worktree isolation worked as intended. This Tide ran in a dedicated git worktree created via `tide-worktree create laplace grammar-j-function` (the K1 helper landed earlier today). Four other Tide branches were active concurrently in their own worktrees; no Laplace.lean import-line clobber, no stray-message commits, no branch-toggle file disappearances. The worktree discipline demonstrably solves the failure class the 6 May cross-susc-deriv-harmonic retrospective (Tide 13) flagged as ‘‘overdue.’’

5 What was Mathlib, what was new

Mathlib provided. `MeasureTheory.tendsto_integral_filter_of_dominated_convergence` for the DCT step². `MeasureTheory.Measure.integral.comp_mul_right` for the change of variables (with the Jacobian factor $|a^{-1}|$). `tendsto_rpow_atTop` (with the Real-namespace gotcha above), `Tendsto.div_atTop`, `Tendsto.const_mul`, `Tendsto.sub_const` for the pointwise-limit chain. `Real.rpow_natCast`, `Real.rpow_mul`, `Real.mul_rpow` for the $t^{1/4}$ bookkeeping. `MeasureTheory.integral.add`, `integral_mul_const` for the corollary’s split.

Tide 7 (laplace seabed) provided. `quartic_partition` (the closed form $\int e^{-tw^4/24} dw = \frac{1}{2}(24/t)^{1/4}\Gamma(1/4)$) and `quartic_integrable_pow` (envelope integrability for the DCT bound) are the only seabed pieces used.

What was new. The substitution lemma `jfunction_centered_subst` --- a $t^{1/4}$ -prefixed change of variables that turns the original integrand into a t -independent shape. The dominator-integrability helper. The two headline theorems. 280 lines total.

Out of scope. The generic- k version (for $K = w^{2k}/(2k)!$ at $k = 1, 2, 3, \dots$). The hypothesis weakening to `AEMeasurable + ContinuousAt 0`. The asymptotic-equivalence form $J(t) \sim \pi(0)Z(t)$ (rather than the `Tendsto` form), which is one `Tendsto.div`-style step from what we have.

6 Lessons

- *Reshape the target if GPT proposes a smaller one.* The deliberation’s headline `reshape` --- from the direct form (Candidate A) to the centred form (X) --- was load-bearing. A direct-form proof would have had the $(1/2) \cdot 24^{1/4} \cdot \Gamma(1/4)$ constant threading through the DCT step; the centred form puts ‘‘zero on the right’’ in the headline and lets the constant fall out by `quartic_partition` in

²The `atTop` : `Filter ℝ` is countably generated, so the `IsCountablyGenerated` typeclass slot is filled automatically.

a clean corollary. Cleaner statement, cleaner proof, smaller line count. GPT correctly identified this as a reshape worth the rename.

- *The K1 worktree discipline pays off immediately.* Today this Tide ran with four other concurrent Tide branches active. Zero collisions, zero file disappearances, zero stray-message commits. The K1 wrapper is a structural fix to the failure class documented in two prior retrospectives; this is the first Tide that ran fully under it. Worth promoting in the Tide skill text from ‘‘recommended’’ to ‘‘mandatory’’ (already done).
- *Use Continuous when AEMeasurable costs a Tide-substantial detour.* The deliberation correctly identified the mathematically-minimal hypothesis class. The formalisation correctly identified that the weakening is a follow-up, not a Tide blocker. Both are right; the discipline is to write *both* into the tide log so the deliberation’s intent isn’t lost.
- *Three new CLAUDE.md entries.* The ‘ π ’-as-binder issue, the namespace inconsistency on `tendsto.rpow.atTop`, and the `.symm` dot-notation gotcha are all small but recurrent friction points. Worth landing.

7 Follow-ups

- *Hypothesis weakening to AEMeasurable + ContinuousAt 0.* The deliberation’s chosen minimum. Requires `AEMeasurable.comp.quasiMeasurePreserving` for the composition $\pi \circ (\cdot/a)$ plus a parallel argument inside the DCT bound step. 50 lines.
- *Generic- k J-function for $K(w) = w^{2k}/(2k)!$.* Mirror today’s proof for $k = 3$ (sextic, using `Sextic.lean`’s `sextic_partition`) and abstract a generic- k template. The `Real.rpow` exponent bookkeeping for $t^{1/(2k)}$ is the main friction. Estimated 250 lines as a follow-on Tide. Pairs with the pending A1 (generic even-monomial 1D template) on the May 7 candidates survey.
- *Asymptotic-equivalence form $J(t) \sim \pi(0)Z(t)$.* The grammar paper’s actual statement is in equivalence form, not `Tendsto-times-power` form. Conversion is one `Tendsto.div`-style step away (`Tendsto (J(t)/Z(t)) atTop (N (pi0)))onceZ(t)iswrappedasacontinuousnonzerofunctionoft.` 10lines.
- *Promote the substitution lemma to lean-common.* The pattern $t^{1/(2k)} \cdot \int f(t) \cdot w^{2k}/(2k)! dw = \int f(u^{2k}/(2k)!) du$ for any locally bounded f is generic to single-monomial Laplace-method computations. Worth lifting to `lean-common` once a second Laplace-method tide uses it.
- *\leanref-tag the grammar paper.* Once the grammar paper has a clear blueprint pointing at this identity, tag with `\leanref` pinned to commit 5d2ae2c. Process tide.
- *Mark E4 Closed in the May 7 candidates survey.* SRI-side close-out per the Tide skill’s Step-4 discipline.

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GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2, including the load-bearing reshape from Candidate A (direct form) to Candidate X (centred form), the warning about the constant in the generic- k version ($1/k$, not $1/(2k)$), and the recommendation against an `IsLittle0` reformulation. The full consult is preserved at `projects/primer/tide-log/gpt55_tide_grammar_j_function.v1.md`.